

# Vendure

## Documentation Infrastructure Audit Report

Prepared by VeriOps

2026-04-29 11:19 UTC

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**Risk: Critical**

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### Key Metrics

Pages scanned: 1383

Report type: Premium Executive Audit

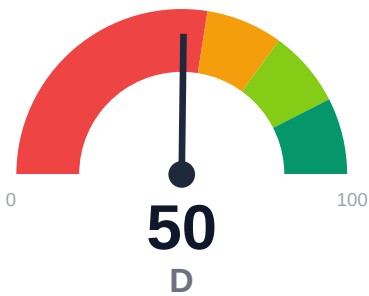
Methodology: 5-pillar automated analysis + expert synthesis

### Headline Findings

- Broken internal links: 30
- SEO/GEO issue rate is high: 31.56%
- API usage-doc coverage is low: 21.95% (1212 API pages detected)
- Example reliability: 0% on 1383 pages (likely detection limitation). Site may use JS-rendered code blocks.

This report is generated automatically by the VeriOps platform. Estimates are directional.

## Executive Summary



Audit of docs.vendure.io covered 1383 pages with 100% crawl coverage and 91.5% overall confidence. Documentation contains 30 confirmed broken internal links requiring immediate remediation. API reference coverage is critically low at 21.95% despite 1212 API pages detected, indicating significant gaps in usage documentation. SEO/geo issues affect 31.56% of pages. Example code detection reported 0% reliability across all pages, flagged as a likely technical detection limitation due to JavaScript-rendered content.

External posture: SEO/GEO issue rate **31.6%**, public API coverage **21.9%**, broken links **30**.

|                              |                           |                                |
|------------------------------|---------------------------|--------------------------------|
| <b>50</b><br>Audit Score     | <b>D</b><br>Grade         | <b>Critical</b><br>Risk Band   |
| <b>1383</b><br>Pages Scanned | <b>30</b><br>Broken Links | <b>31.6%</b><br>SEO Issue Rate |

### Per-Site Breakdown

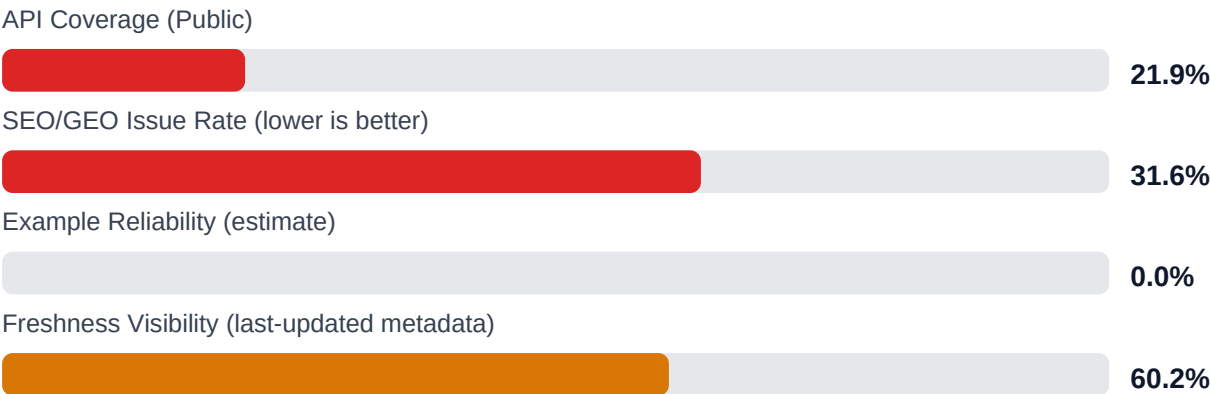
| Site URL                 | Pages | Broken Links | API Cov % | Example Rel % | SEO Issue % |
|--------------------------|-------|--------------|-----------|---------------|-------------|
| https://docs.vendure.io/ | 1383  | 30           | 21.9      | 0.0           | 31.6        |

## Industry Benchmark Comparison

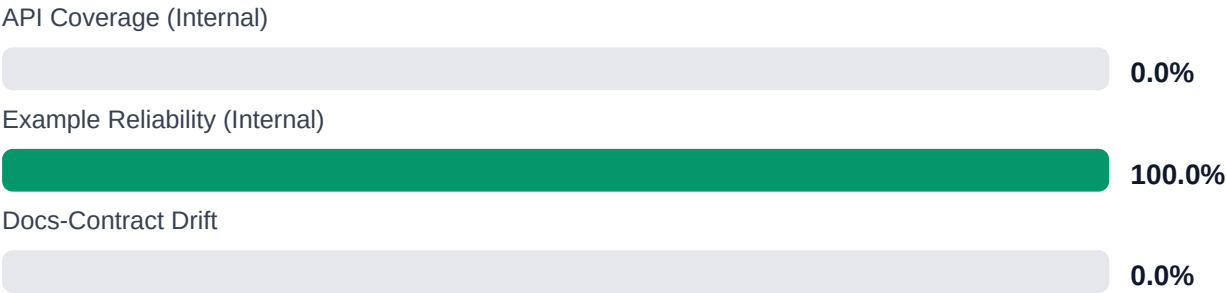
| Benchmark                         | Score | Gap |
|-----------------------------------|-------|-----|
| Top performers (Stripe, Twilio)   | 92+   | -42 |
| Industry average (developer docs) | 85    | -35 |
| Minimum acceptable                | 70    | -20 |
| Your score                        | 50    | -   |

Gap to industry average: **35 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

## Board-Level Metrics



### Internal Metrics (from scorecard)



## Financial Exposure Model

| Low Estimate                                                                                                                                        | Base Estimate                                                                                                                                        | High Estimate                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>\$140,822</div> <div>annual exposure</div> <div>Remediation: \$2,090</div> <div>Monthly loss: \$4,125</div> <div>Opportunity: \$7,436/mo</div> | <div>\$249,513</div> <div>annual exposure</div> <div>Remediation: \$5,445</div> <div>Monthly loss: \$12,903</div> <div>Opportunity: \$7,436/mo</div> | <div>\$431,728</div> <div>annual exposure</div> <div>Remediation: \$8,800</div> <div>Monthly loss: \$27,808</div> <div>Opportunity: \$7,436/mo</div> |

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

| ID                     | Issue                                  | Severity | Monthly Loss | Fix Cost |
|------------------------|----------------------------------------|----------|--------------|----------|
| F-LAYERS               | Missing required doc layers            | HIGH     | \$1,782      | \$825    |
|                        |                                        |          |              |          |
| F-FRESHNESS            | Documentation freshness debt           | MEDIUM   | \$1,122      | \$550    |
|                        |                                        |          |              |          |
| F-RETRIEVAL            | RAG retrieval quality risk             | MEDIUM   | \$2,257      | \$990    |
|                        |                                        |          |              |          |
| F-API-COVERAGE         | Undocumented API operations            | LOW      | \$2,904      | \$1,100  |
|                        |                                        |          |              |          |
| F-EXAMPLES-RELIABILITY | Code examples fail or are non-runnable | LOW      | \$2,244      | \$935    |
|                        |                                        |          |              |          |
| F-DRIFT                | Code/docs contract drift               | LOW      | \$1,901      | \$660    |
|                        |                                        |          |              |          |
| F-TERMINOLOGY          | Terminology inconsistency              | LOW      | \$693        | \$385    |
|                        |                                        |          |              |          |

Priority Action Plan (Next 14 Days)

| ID                     | Finding                                | Severity |
|------------------------|----------------------------------------|----------|
| F-LAYERS               | Missing required doc layers            | HIGH     |
|                        |                                        |          |
| F-FRESHNESS            | Documentation freshness debt           | MEDIUM   |
|                        |                                        |          |
| F-RETRIEVAL            | RAG retrieval quality risk             | MEDIUM   |
|                        |                                        |          |
| F-API-COVERAGE         | Undocumented API operations            | LOW      |
|                        |                                        |          |
| F-EXAMPLES-RELIABILITY | Code examples fail or are non-runnable | LOW      |
|                        |                                        |          |

Recommended Actions

1. Fix all 30 broken internal documentation links immediately [impact: critical, effort: low]
2. Conduct manual audit of API usage documentation to address 78% coverage gap and create remediation plan for top 20 API endpoints [impact: critical, effort: high]

3. Resolve SEO/geo issues on 436+ affected pages (missing meta descriptions, titles, or geo-targeting problems) [impact: high, effort: medium]
4. Implement server-side rendering or static generation for code examples to enable automated validation [impact: high, effort: high]
5. Add last-updated metadata to remaining 40% of pages and establish content review cadence [impact: medium, effort: medium]

## Expert Analysis: Strengths and Risks

### Strengths

- + Complete crawl coverage achieved (100%) with 1383 pages successfully indexed
- + High overall audit confidence at 91.5% with strong link verification (95%) and crawl confidence (95%)
- + Multi-project topology successfully mapped with no repository navigation link breakage
- + Freshness metadata present on 60.23% of pages, enabling content age tracking
- + Zero unverified links, indicating thorough link validation process completed

### Risks

- 30 confirmed broken internal documentation links create poor user experience and damage credibility
- API reference coverage at 21.95% leaves 78% of API surface without adequate usage documentation
- SEO/geo issues on 31.56% of pages (436+ pages) harm discoverability and search rankings
- Example code detection failure (0%) prevents validation of code sample accuracy and runability
- Nearly 40% of pages lack freshness metadata, obscuring content staleness risk
- 1212 API pages detected but low coverage suggests systematic gap in tutorials, guides, or integration examples
- JavaScript-rendered code blocks may hide broken examples from users and internal quality processes

### Limitations

- \* External audit cannot verify accuracy of technical content, code correctness, or conceptual explanations
- \* JavaScript-rendered code examples (0% detection) were not evaluated for syntax, runability, or version compatibility
- \* API coverage metric measures reference-to-usage linking only; does not assess documentation quality, completeness, or clarity
- \* Freshness metadata presence does not guarantee content is actually current or reflects latest product version
- \* Audit cannot detect broken external links to third-party services, dependencies, or partner documentation

## Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

### 1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

### 2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

### 3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

### 4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

### 5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

| Assumption                          | Value |
|-------------------------------------|-------|
| engineer_hourly_usd                 | 110.0 |
| support_hourly_usd                  | 50.0  |
| release_count_per_month             | 8.0   |
| baseline_manual_sync_hours_per_week | 12.0  |
| avg_release_delay_hours             | 4.5   |
| monthly_support_tickets             | 800.0 |
| docs_related_ticket_share           | 0.22  |
| avg_ticket_handling_minutes         | 24.0  |

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

| Finding                                                          | Evidence Source                                   | Confidence |
|------------------------------------------------------------------|---------------------------------------------------|------------|
| F-LAYERS -- Missing required doc layers                          | policy pack + frontmatter layer scan              | HIGH       |
| F-FRESHNESS -- Documentation freshness debt                      | frontmatter date scan                             | HIGH       |
| F-RETRIEVAL -- RAG retrieval quality risk                        | retrieval_evals_report.json                       | HIGH       |
| F-API-COVERAGE -- Undocumented API operations                    | OpenAPI + docs text scan                          | HIGH       |
| F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable | examples_smoke_report.json                        | HIGH       |
| F-DRIFT -- Code/docs contract drift                              | pr_docs_contract.json + api_sdk_drift_report.json | HIGH       |
| F-TERMINOLOGY -- Terminology inconsistency                       | glossary.yml forbidden terms scan                 | HIGH       |

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.